MsC Fundamental Principles of Data Science
TFM Title

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Abstract

In this research, a laminar steady flow conducted throughout a cylindrical pipe is generated using a fan. Using the experimental data obtained measuring simultaneously three observation points and a data set simulated using COMSOL, the Ensemble Kalman Filter (EnKF) is implemented as the optimal data assimilation method to improve the accuracy of a fitted model for the configuration studied.

The experimental and simulated distributions of velocity over time are treated as time series, given the dependency on the past values of the velocity for each point. An introduction to state-space theory is presented to fully understand the fundamentals of the sequential method presented by the Kalman recursions. Subsequently, fluid flow formulation is established using the principles of fluid dynamics as described by the Navier-Stokes and the continuity differential equations. The corresponding assumptions on the boundary and initial conditions are considered in order to find a fitting system to the experimental real setup. Given the complexity of finding an analytical exact solution for the experimental system, taking into account the real assumptions that deviate from a regular Poiseuille flow, the DA methods are presented and studied as the optimal approaches in order to find the best estimates for the true state vector of the velocity profile.

Chapter 1

Theoretical introduction/State of the Art

1.1 Sktime paper

Chapter 2

Methodology

Chapter 3

Implementation

Chapter 4

Conclusions

The KF and EnKF are emerging as the optimal DA methods in CFD simulation problems applied to real world experimental systems. In this research, both DA algorithms have been used to forecast, inference and interpolate the velocity distribution of a turbulent wind flow inside a round pipe. Using two different experimental points and the initial velocity measurement, the DA methods have been successfully carried out, evincing the effectiveness of the introduced update step incorporating the experimental data to the inference algorithm.

Using the punctual distributions to implement both the KF and the EnKF, their accuracy for approximating the experimental distribution is manifested when analysing the resulting filtering distributions. The robustness of the regular KF filtering distributions is outstanding when adjusting the velocity distribution. This results matches the theoretical expectations, since the experimental configuration is not a complex high-dimensional model. Therefore, the EnKF, although more effective than the simulated CFD data, provides an approximation for the KF filtering distribution, given the algorithm structure, based on implementing the recursion over an ensemble from the whole distribution.
The forecast problem, applied over the KF for future time steps has provided successful results, approximating the true state as accurately as the EnKF adjusts the experimental distribution used in the algorithm as new observations.
On the other hand, the interpolated filtering distribution provides improvable results. Note that the EnKF key idea is the use of ensemble data sets, providing a longitudinal and temporal filtering state, defined for every position node and every time step. However, due to the low number of the punctual incoming data, the velocity profile is not precisely represented by only two points. Therefore, the filtering distributions adjusted closer to the measurement points provide better approximations of the true state than the points located at distant positions. In order to manifest the EnKF real potential, a more complex experimental configuration is necessary.
In conclusion, despite the difficulties drawn from the limited experimental distribution, the efficacy of the DA approaches presented is outlined and proven over forecast and approximation problems. Its use for more complex fluid systems, such as Kalman vortex flows or high-dimensional state-space models, and the implementation of the corresponding algorithm with more measurement points may be considered in future work, motivated by the improvement of the results and the aim to evidence the real potential of the DA algorithms.

Acknowledgements

Appendices

Appendix A

Sktime

A.1 SKTIME: A unified interface for Machine Learning with time series

By definition, [2], a *time series* is a set of observations Y_t, each one being recorded at specific time t. In this particular research, we apply the theory for *discrete-time time series* given that the set T_0 of times at which observations are made is a discrete set as the observations are made at fixed time intervals, one per second.
Time series are particularly characterised by the statistical dependency among different time points. Due to this dependency, time series data does not naturally fit into the standard ML framework for tabular (or cross-sectional) data, which implicitely assumes observations to be independent and identically distributed (i.i.d).

When studying time series data, we can have single (or univariate) time series or multiple time series data. In the case we have diverse observations over time in our data set we can distinguish two different cases:

1. Multivariate time series data, where two or more variables are observed over time, with variables representing different kinds of measurements within a single experimental unit;

2. Panel data (longitudinal data), where multiple independent instances of the same kinds of measurement are observed.

Sktime, [12], is a innovative Python package compatible with the scikit-learn package, developed in order to implement Machine Learning models with time series. There is a very broad set of ML problems using time series. The most common tasks that can be studied throughout time series ML are detailed as follows:

1. **Time series regression/classification**: we observe N i.i.d. panel data training instances of feature-label pairs $(\boldsymbol{x}_{,i}, y_i)$, with $\boldsymbol{x}_i = (x_i(t_1), \ldots, x_i(t_T))$. Similar to the regular ML regression tasks, the training data is used to learn a predictor $\hat{f}$ that can accurately predict a new target value, such that $\hat{y} = \hat{f}(\boldsymbol{x_j})$, for a new input unseen time series $\boldsymbol{x}_j$. For regression, $y_i \in \mathbb{R}$, while for classification y_i takes values from a finite discrete set of class values.

2. **Classical forecasting:** Given past observations $\boldsymbol{x}_i = (x_i(t_1), \ldots, x_i(t_T))$ of a single time series, the task consists on learning a forecaster $\hat{f}$ which can make accurate temporal forward predictions $\hat{x} = \hat{f}(s_j)$ of observations at given time points s_j of the forecasting horizon.

3. **Supervised/panel forecasting:** We observe N i.i.d panel data training instances $\boldsymbol{x}_i$, $i = 1, \ldots, N$, with $\boldsymbol{x}_i = (x_i(t_1), \ldots, x_i(t_T))$. this problem is based on using the training data to learn a supervised forecaster $\hat{f}$ that can make accurate temporal forward predictions $\hat{x}_i = \hat{f}(\boldsymbol{x}_j, x_j)$ for a new instance $\boldsymbol{x}_j$ at given time steps $s_1, \ldots, s_H$.

4. **Time series annotation**: Given observations $\boldsymbol{x} = (x(t_1, \ldots, x(t_T))$ of a single time series, the task is to learn an annotation that accurately predicts a series of annotations $\hat{\boldsymbol{y}} = (\hat{y}(a_1), \ldots, \hat{y}(a_A))$ for the observed series, where a_i denotes the time indices of the annotations. [1]

A general approach to treat time series in ML is considering a **reduction** of these tasks, which consists on decomposing the initial problem into simpler tasks that can be composed to give a solution to the original task. This approach is able to transform an initial task of time series forecasting or classification into a tabular regression/classification.
However, are we really interested on considering this kind of approach?
Maybe, our train of thought or our *modus operandi* may be more focused on having an initial time series forecasting/classification and treating the data as the corresponding non-tabular information, in order to generate an efficient approach that will not need to convert or decompose the original task into multiple problems just to obtain a classical regression/classification with tabular data. This may save time and would be a direct approach, instead of having to think on the decomposition path and iterativelly solving diverse ML tasks instead of one solid problem. Nonetheless, this point of view presents other limitations, specially regarding the available tools, since the treatment of time series data is not optimised. We may firstly focus on this initial analysis, corresponding to the pre-processing when facing a time series ML problem.

Time series analysis is focused on finding the distribution of the time dependent random variables given as the input information. We may refer to this distribution as the input distribution. In order to be more familiarized with time series distributions we may consider some theoretical introduction.
It is remarkable to highlight that this field of ML tasks is still developing and there are some limitations concerning the possible models we can study and implement using these open-source machine learning packages.
Therefore, the corresponding package has been developed and is still being improved in order to enable a concise and robust time series analysis and successful implementation of the aforementioned or further time series ML models.

A.1.1 API design

The main goal of `sktime` is to create a unified API for multiple time series tasks, extending `scikit-learn` interface to the temporal setting, while staying close to its syntax and logic.

Time series representation using this package differs from the general representation of tabular data provided by `pandas` because it allows the user to represent time series data in a

[1] For further information on annotation problems, and even some theory and state of the art of time series ML problems, see [1].

nested format, with rows representing i.i.d. instances and columns representing different variables of the model, however, the cells now no longer represent only primitives but also entire time series. This representation's benefits relays on the fact of being able to handle multivariate, panel and time-heterogeneous data[2].

The **task-specific estimators** are consistent with the `scikit-learn` terminology, consisting of the model used from training data and a predict method used for making the predictions based on the fitted model.
Similar to estimators, **transformers** have a uniform API consisting of a fit and a transform used to transform the input data, and a corresponding method for the inverse transformation if available, in addition to the common hyper-parameters interface. We classify the different types of transformations into:

1. **Tabular:** `scikit-learn` like transformers,
2. **Series-to-primitves**
3. **Series-to-series**: operates over time points and transforms time series into a different time series (e.g. series of fitted auto-regressive coefficients)
4. **Detrending:** Operates over time points and transforms given time series, returning a detrended time series in the same domain as the input (Why isn't this an specific case of the previous type? It actually gets a time series and provides a time series. For panel data, it can be applied like a series-to-series transformation by iterating over instances.). **In forecasting, it is fitted only during training.**

A.2 Project motivation

`sktime` package has multiple forecasters (`ThetaModel()`, or `autoARIMA` models) that in order to compute the predictions require the setting of several hyperparameters. When facing seasonal time series, the **detrending** transformation is computed over the original time series on the training process, as aforementioned. In those cases, one of the given hyperparameters of the `fit` process is `sp`, corresponding to the period of the seasonal time series, i.e. the number of observations in the periodic interval. In practice, this number is unknown except for the cases in which the characterisation of the original model can help inferring the intuitive periodicity (for example, yearly or monthly periodicity in stocks or regular time series). Therefore, in most cases trial and error approaches are followed in order to determine the best estimate of this parameters. This could be computed by setting different values for the `sp` parameter and computing the forecast in order to compare multiple results in terms of different metrics criterion over the predictions.

For each value of `sp` the model computationally requires a convolution of increasing length, which can be found on the class `Deseasonalizer`. Within this class the class `seasonal_decompose` is called which in turn calls `convolution_filter` (which, more precisely uses `signal.convolve(x, filt, mode='valid'`.

When deseasonalizing a time series, we are therefore considering a convolution, which in terms of the time series analysis corresponds to applying a filtering, in particular, a moving average on a sliding window across the whole time domain of the time series.

[2]Find proper definition of this concept

More explicitly, for a time series, X_i, with period T, some suitable filters used to identify seasonality are given as follows

$$A_q(X_i) := \frac{1}{2q+1} \sum_{j=-q}^{q} x_{i+j}$$

when we have odd period $T := 2q + 1$, and

$$A_q(X_i) := \frac{1}{2q} \left(\frac{1}{2} X_{i-q} + X_{i-q+1} + \cdots + X_{i+q-1} + \frac{1}{2} X_{i+q} \right),$$

for even period, $T = 2q$.
Nonetheless, theoretically, this deseasonalizing can be computed either by applying the aforementioned filter on the time series or either by using differentiation of the time series, understanding this differentiation by applying the Backward operator, B[3], in a smart way depending on the characterisation of the time series itself. This is, if we want to subtract a periodicity of period T from the original time series $\{X_t\}$ we would consider the differentiator operator, defined as follows:

$$\nabla_T X_t := (I - B^T) X_t = X_t - X_{t-T}$$

Therefore, when having a seasonal time series, i.e. $\boldsymbol{X}_t = \boldsymbol{X}_{t-T} + \varepsilon_t$, where $\varepsilon_t \sim WN(0, \sigma^2)$[4], in order to deseasonalize this time series we would apply the corresponding differentiation operator previously defined:

$$\nabla_T \boldsymbol{X}_t = (I - B^T) \boldsymbol{X}_t = \boldsymbol{X}_t - \boldsymbol{X}_{t-T} = \varepsilon_t \sim WN(0, \sigma^2).$$

Observe however that this convolution filter of dimension `sp` implies a high computational cost increasing as the hyperparameter grows bigger. Therefore, finding a more efficient way of computing this fitting might yield to lower training time.

A.2.1 Proposal of the project: Build a forecaster in which `sp` would not be an hyperparameter but instead it would be determined within the fitting process in an efficient approach

For the initial proposal statement, odd periods are considered. However, in the case of even periods, similar approach can be followed, but considering the slightly modifications on the definition of the convolution. Additionally, a `ThetaModel` is used as the forecaster studied, however interesting further studies can be considered with ARMA and diverse forecasters of the package `sktime`.

Continuation on Notebook for the moment.

Proves d'implementació, no acaba de quadrar mirar millor!

```
def conv_signal_fast(i, integral_image, filter_value, n_filter):
    '''
    Filter as implemented for the odd period
```

[3]The backward operator corresponds to considering the previous time step on the time series notation, i.e. $BX_t = X_{t-1}$

[4]This corresponds to a white noise distribution, this is, the random variables ε_t have mean 0, constant variance σ^2 and are uncorrelated, i.e. $EX_tX_s = 0$ for any $s \neq t$

```
    '''
    if i>0:
        return filter_value*(integral_image[i+n_filter-1] - integral_image[i-1])
    else:
        return filter_value*integral_image[i+n_filter-1]

def conv_signal_fast_e(i, integral_image, filter_value, n_filter, y_train_np):
    '''
    Filter for even period using the y_original
    '''
    if i>0:
        return filter_value*(integral_image[i+n_filter-1] - integral_image[i-1+1] + 1/2*(y_train_np[i-1] + y_train_np[i+n_filter]))
    else:
        return filter_value*(integral_image[i+n_filter] + 1/2*y_train_np[i+n_filter])

def conv_signal_fast_e2(i, integral_image, filter_value, n_filter):
    '''
    Deduced expression with i-1
    '''
    if i>0:
        return filter_value*1/2*(integral_image[i+n_filter-1] + integral_image[i+n_filter] - 2*integral_image[i] + integral_image[i-1] -integral_image[i-2])
        #return filter_value*1/2*(integral_image[i] - integral_image[i-1] + integral_image[i + n_filter] - integral_image[i+n_filter-1] - 2*integral_image[i+1])
    elif i == 0:
        #return filter_value*1/2*(integral_image[i+n_filter] - integral_image[i+n_filter-1] - 2*integral_image[i+1] + integral_image[i])
        return filter_value*1/2*(integral_image[i+n_filter] +  integral_image[i+n_filter-1])
    elif i == 1:
        return filter_value*1/2*(integral_image[i+n_filter] +  integral_image[i+n_filter-1])

def conv_signal_fast_e3(i, integral_image, filter_value, n_filter):
    '''
    Deduced expression with i
    '''
    if i>1:
        #return filter_value*1/2*(integral_image[i+n_filter] - integral_image[i+n_filter-2] - integral_image[i] + integral_image[i-2])
        return filter_value*1/2*(integral_image[i] - integral_image[i-1] + integral_image[i + n_filter] + integral_image[i+n_filter+1] - 2*integral_image[i+1])
    elif i == 0:
        return filter_value*1/2*(integral_image[i+n_filter] +  integral_image[i+n_filter+1]) # + 2*integral_image[i+1])
    elif i == 1:
        return filter_value*1/2*(integral_image[i+n_filter] +  integral_image[i+n_filter+1])
        #return filter_value*1/2*(integral_image[i+n_filter] +  integral_image[i+n_filter-2])
```

```
i = 0
print(np.sum(y_train_np[i:len(filt)+i]*filt))
#print(conv_signal[i])
filter_value = filt[1]
n_filter = period
#period # len(filt) #
```

```
print(conv_signal_fast(i, integral_image, filter_value, n_filter))
print(conv_signal_fast_e(i, integral_image, filter_value, n_filter, y_train_np))
print(conv_signal_fast_e2(i, integral_image, filter_value, n_filter))
print(conv_signal_fast_e3(i, integral_image, filter_value, n_filter))

print('-----------------------------------------------------')
n_filter = period+1
print(conv_signal_fast(i, integral_image, filter_value, n_filter))
print(conv_signal_fast_e(i, integral_image, filter_value, n_filter, y_train_np))
print(conv_signal_fast_e2(i, integral_image, filter_value, n_filter))
print(conv_signal_fast_e3(i, integral_image, filter_value, n_filter))
```

A.3 Theoretical ideas over trend identification using same approach

Note that, when consider for example autoregressive time series models (AR(p) part of ARMAs, for example) we seek to find the order of the AR(p) model in order to obtain a white noise. Therefore, a similar approach to when we consider periodical time series would be conducted, this time applying a slightly modified differentiation.
For example, for a time series following an AR(1) model, this is $X_t = \phi X_{t-1} + \varepsilon_t$, where $\varepsilon_t \sim WN(0, \sigma^2)$, and $|\phi| < 1$, then we can subtract this autoregression by applying

$$(I - \phi B)X_t = X_t - \phi X_{t-1} = \varepsilon_t \sim WN(0, \sigma^2)$$

Similarly, this can be applied to random walks, for example $X_t = X_{t-1} + \varepsilon_t$, by considering $\nabla X_t = \varepsilon_t$.
Furthermore, this could even be applied to time series models with higher order of dependency of previous lags, for example $X_t = X_{t-2} + \varepsilon_t$, by considering $\nabla^2 X_t = (I - B)^2 X_t$.
Let's take one more step into this consideration. Assume we have a seasonal and autoregressive model (SARMA(s, p, q), without the moving average, although we can also get rid of this, think more!). Fore example, consider a $X_t \sim SARMA(T, 1, 0)$, which would be equivalent to a seasonal random walk. Therefore, we could fully decompose it and get a white noise from it by considering the following operations:

$$\nabla_T \nabla X_t = (I - B^T)(I - B)X_t = \varepsilon_t \sim WN(0, \sigma^2)$$

Therefore, by applying twice the differentiation method we can first discard the stochastic trend given by the random walk, and hence deseasonalize the remaining time series, in order to obtain a plain white noise again.

Adding all this up, it is direct to see that there is a potentially general approach that we can follow in order to fully decompose a given time series, in order to find its trend, autoregressive behaviour, as well as the seasonality it follows. If we are able to recognise the equivalency on the application of our improved filter (the moving average filter, i.e. the convolution over a sliding window) with the theoretical differentiation approach of removing seasonality, and trends, we could be able to apply our model practically to fully decompose a time series, and therefore construct a robust pre-processing of the input distribution.

Appendix B

Theoretical introduction Time Series TFG

B.1 Data Assimilation

Data Assimilation encompasses a group of statistical methods for feedback of experimental values to numerical simulation to improve accuracy. In this paper, data assimilation is used to provide a relevant mathematical framework to combine the experimental observations with computational fluid dynamics (CFD) predictions in order to obtain a better description of the model inputs like initial conditions, boundary conditions or model parameters, based on the methodology approached in articles [3], [10], [13]. In data assimilation, the error term of the numerical simulation is represented by a probability. Then, the numerical simulation is treated as a system model. When the experimental data is given, the probability distribution of error is modified to reduce uncertainty of the numerical simulation. This process also improves the accuracy of numerical simulation, [10].

The purpose of DA is to determine the best possible forecast state using new real time observations and short range predictions. Intuitively, an initial forecast is modelled, using the governing equations of the physics system. Using this initial distribution, the next-step variable is predicted with its corresponding covariance matrix, estimating the forecast associated error. The groundbreaking idea surrounding the DA approach is an additional update step that uses new real world observations on an experimental system, to improve the predicted values and obtain an analysed state that approximates more accurately the distribution of the real variable, called the true state. The improved predicted state is used on the subsequent analysis time step along with new real time observations. The quality of the forecasts depends on the approaches taken when using the new information and its uncertainty is computed using the incoming new observations and the corresponding model estimates.

There are two different types of Data Assimilation techniques: variational methods based on optimal control theory and sequential methods based on Bayesian estimation theory. Sequential methods are easier to implement with a flexible adaptation to different models, because they can provide posterior statistics of state variables without changing the intrinsic characteristics of the specific system model structures. In the sequential time-stepping procedure, a previous forecast is compared with newly received observations, the model state is then updated to reflect the observations, a new forecast is initiated and the process iterates repeatedly.

B.1.1 Sequential approach

Data assimilation techniques have been developed mainly in the fields of meteorology and oceanography, focusing in fluid mechanics systems given the progress on CFD techniques becoming able to numerically simulate unsteady flows, such as blades generated flows. DA methods have been performed in order to study 2D steady Reynolds-averaged Navier-Stokes (RANS) equations where the Reynolds stress tensor forms the control vector, [6], and to determine optimal initial and boundary conditions for 2D unsteady direct numerical simulations (DNS), [7].

The sequential DA methods studied and implemented on this research are the regular Kalman Filter (KF) and the Ensemble Kalman Filter (EnKF). The KF was originally presented by [8], which introduces an equation for time evolution of the error covariance matrix. The EnKF was originally introduced by Evensen, [4], introducing improved approaches more orientated to solve nonlinear physical systems with larger number of observations. Both sequential methods have later been further depeloped, implemented and examined in a large number of published papers. Recent studies and reviews on the KF and the EnKF are given from a more theoretical point of view in [5] and [2], providing detailed information on the formulation, interpretation and implementation of the sequential DA method.

Based on the approaches and motivations presented in [3], [10] and [13], in this research, the KF and the EnKF were applied to a turbulent wind experiment as a data assimilation method in order to filter and compute the best estimates for the flow velocity distribution. Firstly, a laminar steady flow conducted throughout a cylindrical pipe is generated using a fan. Using the experimental data obtained measuring simultaneously two observation points and a data set simulated using COMSOL, the KF and the EnKF are implemented as the optimal data assimilation methods to improve the accuracy of a fitted model for the fluid dynamical system studied. Subsequently, an exhaustive comparison and analysis between the experimental, simulated and filtered distributions in order to study the effectiviness of the DA approaches and determine the optimal DA method for the particular fluid system.

B.2 State-Space formulation

The experimental and simulated distributions of velocity over time are treated as time series, given the dependency on the past values of the velocity for each point. By definition, [2], a *time series* is a set of observations Y_t, each one being recorded at specific time t. In this particular research, we apply the theory for *discrete-time time series* given that the set T_0 of times at which observations are made is a discrete set as the observations are made at fixed time intervals, one per second.

Time series models can be expressed using the state-space formulation, leading to a clearer representation of the system's governing equations and providing a set of equations that can be managed in the sequential approaches developed by the DA methods presented in this research. To fully understand the KF and the EnKF recursions and algorithms a self-contained introduction to state-space theory is presented.

The state-space representation is an alternative method used to describe a system's dynamics, in which a set of observations stored in the observation state, $\boldsymbol{Y}_t$ are expressed in terms of a system state, $\boldsymbol{X}_t$, defined by Kalman in 1963, [9].
Formally, a state-space model for a (possibly multivariate) time series $Y_t, t \in \{0, \pm 1, \pm 2, \dots\}$

consists of two equations: the observation equation, that expresses the n-dimensional observation Y_t as a linear function of a k-dimensional state variable plus noise, and the state equation, that determines the state X_{t+1} in terms of the previous state X_t and a noise term[1].

Definition B.2.1. *A time series $\{Y_t\}$ has a state-space representation if there exists a state-space model for $\{Y_t\}$, that consists of the observation and state equations given by:*

$$\boldsymbol{Y}_t = M_t \boldsymbol{X}_t + \boldsymbol{W}_t, \tag{B.1}$$

$$\boldsymbol{X}_{t+1} = F_t \boldsymbol{X}_t + \boldsymbol{V}_t, \tag{B.2}$$

for $t \in \{0, 1, 2, \dots\}$, where $\{\boldsymbol{W}_t\} \sim WN(0, \{R_t\})$, $\{\boldsymbol{V}_t\} \sim WN(0, \{Q_t\})$[2], and $EW_tV_s = 0$ for all s and t. $\{M_t\}$ is a sequence of $(n \times k)$ matrices (observation matrices), and $\{F_t\}$ is a sequence of $(k \times k)$ matrices (prediction matrices)

Observation B.2.2. It follows from the observation equation (B.1) and the state equation (B.2) that $\mathbf{X}_t$ and $\mathbf{Y}_t$ have the functional forms for $t = 2, 3, \dots,$

$$\begin{aligned}
\mathbf{X}_t &= F_{t-1}\mathbf{X}_{t-1} + \mathbf{V}_{t-1} \\
&= F_{t-1}(F_{t-2}\mathbf{X}_{t-2} + \mathbf{V}_{t-2}) + \mathbf{V}_{t-1} \\
&\vdots \\
&= (F_{t-1} \circ \cdots \circ F_1)\mathbf{X}_1 + (F_{t-1} \circ \cdots \circ F_2)\mathbf{V}_1 + \cdots + F_{t-1}\mathbf{V}_{t-2} + \mathbf{V}_{t-1} \\
&= f_t(\mathbf{X}_1, \mathbf{V}_1, \dots, \mathbf{V}_{t-1}),
\end{aligned}$$

where $\circ$ denotes the composition of the linear operators F_k. Furthermore, following the same recursive procedure,

$$\begin{aligned}
\mathbf{Y}_t &= (M_{t-1} \circ \cdots \circ M_1)\mathbf{X}_1 + (M_{t-1} \circ \cdots \circ M_2)\mathbf{V}_1 + \cdots + M_{t-1}\mathbf{V}_{t-2} + \mathbf{V}_{t-1} + \mathbf{W}_t \\
&= g_t(\mathbf{X}_1, \mathbf{V}_1, \dots, \mathbf{V}_{t-1}, \mathbf{W}_t).
\end{aligned}$$

Note that, from the assumptions on the noise terms it is clear that

$$\begin{aligned}
E(\mathbf{W}_t\mathbf{X}_s) = 0, \qquad E(\mathbf{W}_t, \mathbf{Y}_t) = 0, \quad 1 \le s \le t \\
E(\mathbf{V}_t\mathbf{X}_s) = 0, \qquad E(\mathbf{V}_t, \mathbf{Y}_t) = 0, \quad 1 \le s \le t.
\end{aligned}$$

Observation B.2.3. Using the (B.1) and (B.2) we can see that if the sequence $\{\mathbf{X}_1, \mathbf{V}_1, \mathbf{V}_2, \dots\}$ is independent (we have already seen that it is uncorrelated on the Observation B.2.2) then $\{\mathbf{X}_t\}$ has the Markov property, i. e. the distribution of $\mathbf{X}_{t+1}$ given $\mathbf{X}_t, \dots, \mathbf{X}_1$ is the same as the distribution of $\mathbf{X}_{t-1}$ given $\mathbf{X}_t$.

Before considering any further analysis, we may introduce a more natural formulation that includes times series in which the time runs from $-\infty$ to ∞:

Definition B.2.4. *Given a time series $\{\boldsymbol{Y}_t\}$, we consider the space and state equations,*

$$\boldsymbol{Y}_t = M\boldsymbol{X}_t + \boldsymbol{W}_t, \tag{B.3}$$

$$\boldsymbol{X}_{t+1} = F\boldsymbol{X}_t + \boldsymbol{V}_t, \tag{B.4}$$

[1] For any further information on the subsequent concepts see [2].

[2] The notation $\{\mathbf{W}_t\} \sim WN(0, \{R_t\})$ indicates that $\{\mathbf{W}_t\}$ is distributed as a white noice. Therefore, the random vectors $\{\mathbf{W}_t\}$ have mean 0 and $EW_tW_s = R_t$ if $t = s$, and 0 otherwise.

for $t \in \{0, \pm 1, \pm 2, \dots\}$*, where* $\{\boldsymbol{W}_t\} \sim WN(0, R)$*,* $\{\boldsymbol{V}_t\} \sim WN(0, Q)$*, and* $E(\boldsymbol{W}_t \boldsymbol{V}_s) = 0$ *for all* s *and* t*, and* M *(observation matrix) and* F *(prediction matrix) are* $(n \times k)$ *and* $(k \times k)$ *matrices.*

It is clear from the definition that neither $\{\mathbf{X}_t\}$ nor $\{\mathbf{Y}_t\}$ is necessarily stationary. In addition, given a time series, the state-space representation is not unique, since we can find different state-space models that ajust the system. However, there are some cases where we can find the unique stationary solution of the state equation.

Definition B.2.5. *The state equation (B.4) is said to be stable if the matrix* F *has all its eigenvalues in the interior of the unit circle, or equivalently, if* $det(I - Fz) \neq 0$ *for all* $z \in \mathbb{C}$ *such that* $|z| \leq 1$*. In this case, the matrix* F *is said to be stable.*

Proposition B.2.6. *In the stable case, the equations (B.3) have the unique stationary solution given by*

$$\boldsymbol{X}_t = \sum_{j=0}^{\infty} F^j \boldsymbol{V}_{t-j-1}.$$

The corresponding sequence of observations is also stationary:

$$\boldsymbol{Y}_t = \boldsymbol{W}_t + \sum_{j=0}^{\infty} M F^j \boldsymbol{V}_{t-j-1}.$$

Proof. It is enough to extend Observation B.2.2 to the case of $t \to -\infty$, so the recursive procedure keeps going once past $\mathbf{X}_1$, instead of using the initial conditions. Therefore, we obtain an infinite sum that converges as a consequence of the fact that F has all eigenvalues inside the unit circle, and equivalently $F^k \to 0$ as $k \to \infty$. □

Example 1 (ARMA(1,1)). Let $\{Y_t\}$ be the causal and invertible ARMA(1,1) process, given by the equations:

$$Y_t = \phi Y_{t-1} + Z_t + \theta Z_{t-1}, \qquad \{Z_t\} \sim WN(0, \sigma^2) \tag{B.5}$$

In terms of the characteristic polynomials $\Phi(B) = 1 - \phi B$, $\Theta(B) = 1 + \theta$, where B is the backshift operator, $BY_t = Y_{t-1}$, the ARMA(1,1) process can be written as shown:

$$\Phi(B) Y_t = \Theta(B) Z_t.$$

Since the process is causal, $|\phi| < 1$ and the process satisfies:

$$Y_t = \Phi^{-1}(B)\Theta(B) Z_t = \Theta(B)\Phi^{-1}(B) Z_t.$$

If we define $X_t = \Phi^{-1}(B) Z_t$, then we get

$$Y_t = \Theta(B) X_t = X_t + \theta X_{t-1} = \begin{pmatrix} \theta & 1 \end{pmatrix} \begin{pmatrix} X_{t-1} \\ X_t \end{pmatrix} = M \mathbf{X}_t, \tag{B.6}$$

where we have defined $M = \begin{pmatrix} \theta & 1 \end{pmatrix}$ as the observation matrix of the state-space representation, and the state vector $\mathbf{X}_t = \begin{pmatrix} X_{t-1} & X_t \end{pmatrix}^T$.
Note that X_t is defined as an AR(1) process, satisfying the equations:

$$\Phi(B) X_t = Z_t \iff X_t = \phi X_{t-1} + Z_t.$$

Therefore, we can write the state equation in terms of $\boldsymbol{X}_t$:

$$\mathbf{X}_{t+1} = \begin{pmatrix} X_t \\ X_{t+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & \phi \end{pmatrix} \begin{pmatrix} X_{t-1} \\ X_t \end{pmatrix} + \begin{pmatrix} 0 \\ Z_t \end{pmatrix} = F\mathbf{X}_t + \mathbf{Z}_t, \tag{B.7}$$

where F denotes the prediction matrix. Using the MA(∞) representation of X_t we find the initial conditions:

$$\mathbf{X}_1 = \begin{pmatrix} X_0 \\ X_1 \end{pmatrix} = \begin{pmatrix} \sum_{j=0}^{\infty} \phi^j Z_{-j} \\ \sum_{j=0}^{\infty} \phi^j Z_{1-j} \end{pmatrix}. \tag{B.8}$$

Consequently, a state-space representation for $\{Y_t\}$ is given by the observation equation, (B.6), the state equation, (B.7), and the initial conditions of the state vector, (B.8).

However, the state-space representation is not unique. Therefore, we can find another pair of equations that furnish a state-space representation of $\{Y_t\}$ by observing that

$$Y_t = X_t + W_t, \qquad \{W_t\} \sim WN(0, \sigma_w^2) \tag{B.9}$$

$$X_t = \phi_1 X_{t-1} + V_t, \qquad \{V_t\} \sim WN(0, \sigma_v^2) \tag{B.10}$$

is a representation of an ARMA(1,1) process $\{Y_t\}$ given by the general expression (B.5), where

$$\begin{aligned} \phi &= \phi_1, \\ \theta &= \frac{-\sigma_v^2 - (1+\phi_1^2)\sigma_w^2 + \sqrt{\sigma_v^4 + 2(1+\phi_1^2)\sigma_w^2\sigma_v^2 + (1+\phi_1^2)\sigma_w^4}}{2\phi_1\sigma_w^2}. \end{aligned} \tag{B.11}$$

In order to prove that (B.9) and (B.10) give the state-space representation for the ARMA(1,1) process $\{Y_t\}$, first we note that by inserting the state equation (B.10) in the definition of Y_t we get

$$\begin{aligned} Y_t = \phi_1 X_{t-1} + V_t + W_t &= \phi_1 Y_{t-1} + V_t + W_t - \phi_1 W_{t-1}, \\ \iff \quad Y_t - \phi_1 Y_{t-1} &= V_t + W_t - \phi_1 W_{t-1}. \end{aligned}$$

Thus, we just have to see that $U_t = V_t + W_t - \phi_1 W_{t-1}$ defines a MA(1) process. Computing the Wold descomposition and the autocovariance function of U_t we can see that:

$$\frac{1}{\theta} + \theta = -\frac{1}{\phi_1}\left(\frac{\sigma_v^2}{\sigma_w^2} + 1\right) - \phi_1.$$

Rewriting the previous equation as a quadratic equation in θ and solving it with the appropriate assumptions, we get the expression of θ given by (B.11), as we wanted.

Overall, we have another state-space representation, given by (B.9) and (B.10), that can be expressed by the vectorial equations:

$$\begin{aligned} Y_t &= \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} X_{t-1} \\ X_t \end{pmatrix} + W_t = M\mathbf{X}_t + \mathbf{W}_t, \\ \mathbf{X}_{t+1} &= \begin{pmatrix} X_t \\ X_{t+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & \phi \end{pmatrix} \begin{pmatrix} X_{t-1} \\ X_t \end{pmatrix} + \begin{pmatrix} 0 \\ V_t \end{pmatrix} = F\mathbf{X}_t + \mathbf{V}_t. \end{aligned}$$

In both cases we have the same F matrix, with eigenvalues $0, \phi$. Then, they both have a stable state equation and therefore a unique stationary solution, given by Proposition B.2.6.

Another state-space representation for the ARMA(1,1) process can be found if we define the state vector as: $X_t = (Y_t, \quad \theta Z_t)^T$, and therefore

$$Y_t = (1, \quad 0)\begin{pmatrix} Y_t \\ \theta Z_t \end{pmatrix} = M\mathbf{X}_t, \tag{B.12}$$

$$\mathbf{X}_t = \begin{pmatrix} Y_t \\ \theta Z_t \end{pmatrix} = \begin{pmatrix} \phi & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} Y_{t-1} \\ \theta Z_{t-1} \end{pmatrix} + \begin{pmatrix} 1 \\ \theta \end{pmatrix} V_t = F\mathbf{X}_{t-1} + \mathbf{V}_t. \tag{B.13}$$

Note that the first component of $\mathbf{X}_t$ is, according to the state equation, $\phi Y_{t-1} + \theta Z_{t-1} + Z_t$ which is, indeed, Y_t.

Observe that the three possible state-space representations given for an ARMA(1,1) process have the same eigenalues of the prediction matrix. This fact is due to the relationship between the eigenvalues of F and the characteristic polynomial of the process' general expression. □

Example 2. It is interesting to study the state-space representation for some nonstationary time series to clarify that state-space formulation does not necessarily define stationary processes, and hence is a more flexible model.

Consider a random walk with noise, given by the general expression:

$$Y_t = Y_{t-1} + Z_t, \quad \{Z_t\} \sim WN(0, \sigma^2).$$

It is direct to see that a possible state-space representation for this model is given by:

$$\begin{aligned} Y_t &= X_t + W_t, \quad \{V_t\} \sim WN(0, \sigma_w^2) \\ X_t &= X_{t-1} + V_t, \quad \{V_t\} \sim WN(0, \sigma_v^2), \end{aligned}$$

where here we consider $\sigma_v^2 = \sigma^2$.

Contemplate now a more complex case, adding a linear trend with a slope that behaves like a random walk (instead of a deterministic slope),

$$\begin{aligned} Y_t &= Y_{t-1} + B_{t-1} + Z_t, \\ B_t &= B_{t-1} + U_{t-1}, \end{aligned}$$

where $\{U_t\} \sim WN(0, \sigma_u^2)$.
Therefore, we can express this model using the state-space representation using the following equations:

$$\begin{aligned} Y_t &= X_t, \\ X_{t+1} &= X_t + B_t + V_t, \\ B_t &= B_{t-1} + U_{t-1}. \end{aligned}$$

Consequently, if we define the state vector as

$$\mathbf{X}_t = \begin{pmatrix} X_t \\ B_t \end{pmatrix},$$

the state-space vectorial equations are given by

$$Y_t = (1\ 0)\begin{pmatrix} X_t \\ B_t \end{pmatrix} + W_t = M\mathbf{X}_t + \mathbf{W}_t,$$
$$\mathbf{X}_{t+1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} X_t \\ B_t \end{pmatrix} + \begin{pmatrix} V_t \\ U_t \end{pmatrix} = F\mathbf{X}_t + \mathbf{V}_t.$$

□

B.3 Kalman filter

The Kalman filter (KF) is a set of recursion equations for determinating the optimal estimates of the state vector $\mathbf{X}_t$ in terms of the observations $\mathbf{Y}_1, \mathbf{Y}_2, \ldots$ and a random vector $\mathbf{Y}_0$ that is orthogonal to $\mathbf{V}_t$ and $\mathbf{W}_t$ for all $t \geq 1$. Depending on the set of information available, the estimation of $\mathbf{X}_t$ in terms of:

- $\mathbf{Y}_0, \ldots, \mathbf{Y}_{t-1}$ defines the prediction problem,
- $\mathbf{Y}_0, \ldots, \mathbf{Y}_t$ defines the filtering problem,
- $\mathbf{Y}_0, \ldots, \mathbf{Y}_n, (n > t)$ defines the smoothing problem.

Definition B.3.1. *Given the random vector $\boldsymbol{X}_t = (\boldsymbol{X}_{1,t}, \ldots, \boldsymbol{X}_{k,t})^T$, we denote the best linear predictor of $X_{i,t}$ in terms of all the components of the observations $\boldsymbol{Y}_0, \ldots, \boldsymbol{Y}_t$ as $P_t(X_i) := P(X_i|\, \boldsymbol{Y}_0, \ldots, \boldsymbol{Y}_t)$. Therefore, we can denote the best linear predictor random vector by $P_t(\boldsymbol{X}_t) := (P_t(X_1), \ldots, P_t(X_k))$.*

Observation B.3.2. By the definition of the best linear predictor, $\hat{\mathbf{X}}_t = P_{t-1}\mathbf{X}_t$ is the unique random vector lineal on $\mathbf{Y}_0, \ldots, \mathbf{Y}_t$ such that $\left(\mathbf{X}_t - \hat{\mathbf{X}}_t\right) \perp \mathbf{Y}_s$, i. e. $E\left(\mathbf{X}_t - \hat{\mathbf{X}}_t\right)\mathbf{Y}_s^T = 0$, for all $s \in \mathbb{N}$.

Observation B.3.3. Note that if all the components of $\mathbf{X}_t$, $\mathbf{Y}_1, \ldots, \mathbf{Y}_t$ are jointly normally distributed and $\mathbf{Y}_0 = (1, \ldots, 1)^T$, then $P_t(\mathbf{X}_t) = E(\mathbf{X}|\mathbf{Y}_1, \ldots, \mathbf{Y}_t)$, for all $t \geq 1$.

Observation B.3.4. P_t is a linear operator, satisfying that given any $(k \times n)$ matrix A and $\mathbf{X}, \mathbf{Y}$ n-variate random vectors with finite second moments, then

$$P_t(A\mathbf{X}) = AP_t(\mathbf{X}),$$
$$P_t(\mathbf{X} + \mathbf{Y}) = P_t(\mathbf{X}) + P_t(\mathbf{Y}).$$

Observation B.3.5. Given $\mathbf{X}, \mathbf{Y}$ variate random vectors with k and n components, respectively, each with finite second moments, then

$$P_t(\mathbf{X}|\mathbf{Y}) = M\mathbf{Y},$$

where $M = E(\mathbf{X}\mathbf{Y}^T)[E(\mathbf{Y}\mathbf{Y}^T)]^{-1} = \Sigma_{XY}\Sigma_Y^{-1}$ is a $(k \times n)$ matrix, Σ_{XY} is the covariance matrix of $\mathbf{X}$ and $\mathbf{Y}$, and Σ_Y^{-1} is any generalized inverse of Σ_Y (i. e. satisfying that $\Sigma_Y\Sigma_Y^{-1}\Sigma_Y = \Sigma_Y$).

The KF approach seeks to find estimates of the true system state and its uncertainty in the corresponding environment by a type of inference called filtering, which attemps to obtain sequentially the posterior distribution of the state at the current time point based on all observations collected so far. The combination of high dimensionality and nonlinearity motivates the modification of the corresponding algorithm towards an approximate filtering method given by

the EnKF that may be introduced subsequently.

In this framework, the goal is to obtain the filtering distribution of the state at every time point t. In Bayesian terms, this is equivalent to calculate the posterior distribution of $\mathbf{X}_t$ conditioned to the given observations $\mathbf{Y}_0, \ldots, \mathbf{Y}_t$, which we may denote as $\mathbf{X}_{t|t}$. The KF recursive process consists of two steps at every time point: a forecast or prediction step, providing the standard best linear predictor of the state vector, $P_{t-1}(\boldsymbol{X}_t)$ and an update step, which consists on computing the desired analysed estimate, $\mathbf{X}_{t|t}$, incorporating the new observations.

Assuming that the filtering distribution at the previous time step is given by $\mathbf{X}_{t-1|t-1}$ and when the errors are normally distributed, i.e. $\mathbf{X}_{t-1|t-1} \sim N(\boldsymbol{\mu}_{t-1}, \Sigma_{t-1|t-1})$, the forecast step computes the forecast distribution for time t as the one-step predictor $\hat{\mathbf{X}}_t := P_{t-1}(\mathbf{X}_t) \sim N(\hat{\boldsymbol{\mu}}_t, \tilde{\Sigma}_t)$ where $\hat{\boldsymbol{\mu}} := F_t \boldsymbol{\mu}$, and $\tilde{\Sigma}_t := F_t \Sigma_{t-1|t-1} F_t^T + Q_t$.
Note that the observation equation of the state-space model and the external data has not been considered on the computation yet. Therefore, on the update step, the forecast distribution is modified using the new data $\boldsymbol{Y}_t$. The update formula can be easily derived considering the joint distribution of $\boldsymbol{X}_t$ and $\boldsymbol{Y}_t$ conditional on the past data $\boldsymbol{Y}_{1:t-1}$.

Explicitly, the two KF steps are described by the following theorems B.3.6 and B.3.7. Note that Theorem B.3.6 expresses the Kalman prediction equations in terms exclusively of the one-step predictors, leading to a theoretically more interesting but less intuitive expression. An illustrative representation of the KF algorithm is shown in Figure B.1.

Theorem B.3.6 (Kalman Prediction equations)**.** *For the state-space model given by the equations* (B.1) *and* (B.2)*, the one-step predictors* $\hat{\boldsymbol{X}}_t := P_{t-1}(\boldsymbol{X}_t)$ *and their error covariance matrices* $\Sigma_t = E\left[(\boldsymbol{X}_t - \hat{\boldsymbol{X}}_t)(\boldsymbol{X}_t - \hat{\boldsymbol{X}}_t)^T\right]$ *are uniquely determined by the initial conditions*

$$\hat{\boldsymbol{X}}_1 = P(\boldsymbol{X}_1 | \boldsymbol{Y}_0), \qquad \Sigma_1 = E\left[(\boldsymbol{X}_1 - \hat{\boldsymbol{X}}_1)(\boldsymbol{X}_1 - \hat{\boldsymbol{X}}_1)^T\right],$$

and the recursions,

$$\hat{\boldsymbol{X}}_{t+1} = F_t \hat{\boldsymbol{X}}_t + F_t K(\boldsymbol{Y}_t - M_t \hat{\boldsymbol{X}}_t), \tag{B.14}$$

$$\Sigma_{t+1} = F_t (I - K M_t) \Sigma_t F_t^T + Q_t, \tag{B.15}$$

$$K = \Sigma_t M_t^T (M_t \Sigma_t M_t^T + R_t)^{-1}, \tag{B.16}$$

for $t \in \mathbb{N} \setminus \{0\}$*. These equations represent the standard way of writing the analysis or prediction equations for the Kalman filter (KF).*

Proof. We shall define the innovations $\mathbf{I}_t$, given by the initial condition $\mathbf{I}_0 = \mathbf{Y}_0$ and the recursive equations for $t = 1, 2, \ldots,$

$$\begin{aligned} \mathbf{I}_t &= \mathbf{Y}_t - P_{t-1}\mathbf{Y}_t = \mathbf{Y}_t - M_t P_{t-1} \mathbf{X}_t \\ &= \mathbf{Y}_t - M_t \hat{\mathbf{X}}_t = M_t \left(\mathbf{X}_t - \hat{\mathbf{X}}_t\right) + \mathbf{W}_t. \end{aligned}$$

It is direct to see from Observation B.3.2 that the sequence $\{\mathbf{I}_t\}$ is orthogonal. In addition, it is direct to see that every vector of the form $\boldsymbol{Y} = A_1 \boldsymbol{X}_1 + \cdots + A_t \boldsymbol{X}_t$ can be expressed as

$$\boldsymbol{Y} = B_1 \boldsymbol{X}_1 + \cdots + B_{t-1} \boldsymbol{X}_{t-1} + C_t \boldsymbol{I}_t,$$

where $B_1, \ldots, B_{t-1}, C_t$ are matrices that depend on the matrices $A_1, \ldots, A_t$. Using this result and the fact that $E(\boldsymbol{X}_s \boldsymbol{I}_t) = 0$ for all $s < t$, the following relation is established:

$$P_t \mathbf{X}_{t+1} = P_{t-1}(\mathbf{X}_{t+1}) + P(\mathbf{X}_{t+1}|\mathbf{I}_t). \tag{B.17}$$

By virtue of the relation (B.17) and from Observations B.3.4 and B.3.5 it follows that:

$$\begin{aligned} P_t \mathbf{X}_{t+1} = \hat{\mathbf{X}}_{t+1} &= P_{t-1}(\mathbf{X}_{t+1}) + P(\mathbf{X}_{t+1}|\mathbf{I}_t) \\ &= P_{t-1}(F_t \mathbf{X}_t + \mathbf{V}_t) + \Theta_t \Delta_t^{-1} \mathbf{I}_t \\ &= F_t \hat{\mathbf{X}}_t + \Theta_t \Delta_t^{-1} \mathbf{I}_t, \end{aligned}$$

where the computation of the conditional probability is given by the correlation matrices as specified by Observation B.3.5 as follows

$$\begin{aligned} E(\mathbf{X}_{t+1} \mathbf{I}_t^T) &= E\left((F_t \mathbf{X}_t + \mathbf{V}_t) \left(M_t \left(\mathbf{X}_t - \hat{\mathbf{X}}_t \right) + \mathbf{W}_t \right)^T \right) \\ &= E\left(F_t \mathbf{X}_t \left(\mathbf{X}_t - \hat{\mathbf{X}}_t \right)^T M_t^T \right) \\ &= F_t E\left(\mathbf{X}_t \left(\mathbf{X}_t - \hat{\mathbf{X}}_t \right)^T \right) M_t^T = \\ &= F_t \left[E(\mathbf{X}_t \mathbf{X}_t^T) - E(\mathbf{X}_t \hat{\mathbf{X}}_t^T) \right] M_t^T \\ &= F_t \left[E(\mathbf{X}_t \mathbf{X}_t^T) - E(\hat{\mathbf{X}}_t \hat{\mathbf{X}}_t^T) \right] M_t^T \\ &= F_t E\left[(\mathbf{X}_t - \hat{\mathbf{X}}_t)(\mathbf{X}_t - \hat{\mathbf{X}}_t)^T \right] M_t^T = \Theta_t. \end{aligned}$$

In order to compute this expectation, Observation B.3.2 is used to justify that $E(\mathbf{X}_t \hat{\mathbf{X}}_t^T) = E(\hat{\mathbf{X}}_t \hat{\mathbf{X}}_t^T)$. Note that if $E(\mathbf{X}_t - \hat{\mathbf{X}}_t)\mathbf{Y}_s^T = 0$, $\forall s \in \mathbb{N}$, particularly it accomplishes for $s = t$, therefore, it is direct to see that

$$E(\mathbf{X}_t - \hat{\mathbf{X}}_t)\mathbf{Y}_t^T = 0 \iff E\mathbf{X}_t(M_t \mathbf{X}_t + \mathbf{W}_t)^T = E\hat{\mathbf{X}}_t(M_t \mathbf{X}_t + \mathbf{W}_t)^T \iff E(\mathbf{X}_t \mathbf{X}_t^T) = E(\hat{\mathbf{X}}_t \mathbf{X}_t^T).$$

From the definition of the best linear predictor the desired equality is obtained.

Similarly,

$$\begin{aligned} E(\mathbf{I}_t \mathbf{I}_t^T) &= E\left((M_t(\mathbf{X}_t - \hat{\mathbf{X}}_t) + \mathbf{W}_t)(M_t(\mathbf{X}_t - \hat{\mathbf{X}}_t) + \mathbf{W}_t)^T \right) \\ &= E\left(M_t(\mathbf{X}_t - \hat{\mathbf{X}}_t)(\mathbf{X}_t - \hat{\mathbf{X}}_t)^T M_t^T \right) + E(\mathbf{W}_t \mathbf{W}_t^T) \\ &= M_t \Sigma_t M_t^T + R_t = \Delta_t, \end{aligned}$$

where the notation $\Theta_t = F_t \Sigma_t M_t^T$ i $\Delta_t = M_t \Sigma_t M_t^T + R_t$ has been introduced for simplicity.

To verify (B.15) we observe from the definition of Σ_{t+1} that

$$\Sigma_{t+1} = E(\mathbf{X}_{t+1} \mathbf{X}_{t+1}^T) - E(\hat{\mathbf{X}}_{t+1} \hat{\mathbf{X}}_{t+1}^T).$$

Using the expression of the linear predictor we get

$$\begin{aligned} \Sigma_{t+1} &= F_t E(\mathbf{X}_t \mathbf{X}_t^T) F_t^T + Q_t - F_t E(\hat{\mathbf{X}}_t \hat{\mathbf{X}}_t^T) F_t^T - \Theta_t \Delta_t^{-1} \Theta_t^T \\ &= F_t \Sigma_t F_t^T + Q_t - \Theta_t \Delta_t^{-1} \Theta_t^T. \end{aligned}$$

□

Example 3. Consider the random walk plus noise model given by the state-space model studied in Example 2:

$$\begin{aligned} Y_t &= X_t + W_t, \quad \{V_t\} \sim WN(0, \sigma_v^2) \\ X_t &= X_{t-1} + V_t, \quad \{V_t\} \sim WN(0, \sigma_v^2). \end{aligned}$$

Applying the Kalman prediction equations we can compute the linear estimator for the process:

$$\begin{aligned} \hat{Y}_{t+1} &= P_t Y_{t+1} = P_t(X_{t+1} + W_{t+1}) = \hat{X}_{t+1} \\ &= \hat{X}_t + \Theta_t \Delta_t^{-1} I_t \\ &= (1 - a_t)\hat{Y}_t + a_t Y_t, \end{aligned}$$

where $a_t = \Theta_t \Delta_t^{-1} = \frac{\Sigma_t}{\Sigma_t + \sigma_w^2}$. Equivalently, the recursive equations for the correlation matrices are given by

$$\Sigma_{t+1} = \Sigma_t + \sigma_v^2 - \frac{\Sigma_t^2}{\Sigma_t + \sigma_w^2}.$$

□

The Kalman prediction equations lead to an algorithm for recursive calculation of the best linear mean square predictors $P_t \mathbf{Y}_{t+h}$, $h = 1, 2, \ldots$:

$$\begin{aligned} P_t \mathbf{X}_{t+1} &= F_t P_{t-1} \mathbf{X}_t + \Theta_t \Delta_t^{-1} (\mathbf{Y}_t - P_{t-1} \mathbf{Y}_t), \\ P_t \mathbf{X}_{t+h} &= F_{t+h-1} P_t \mathbf{X}_{t+h-1} = \cdots = (F_{t+h-1} \circ F_{t+h-2} \circ \cdots \circ F_{t+1}) P_t \mathbf{X}_{t+1}. \end{aligned} \tag{B.18}$$

Therefore,

$$P_t \mathbf{Y}_{t+h} = G_{t+h} P_t \mathbf{X}_{t+h}. \tag{B.19}$$

Furthermore, from the relation

$$\mathbf{X}_{t+h} - P_t \mathbf{X}_{t+h} = F_{t+h-1}(\mathbf{X}_{t+h-1} - P_t \mathbf{X}_{t+h-1}) + \mathbf{V}_{t+h-1},$$

we find that $\Sigma_{t+h} = E\left[(\mathbf{X}_{t+h} - P_t \mathbf{X}_{t+h})(\mathbf{X}_{t+h} - P_t \mathbf{X}_{t+h})^T\right]$ satisfies the recursions

$$\Sigma_{t+h} = F_{t+h-1} \Sigma_{t+h-1} F_{t+h-1}^T + Q_{t+h-1}.$$

These recursions are remarkably interesting, since they provide the forecast distribution for the state variable, in this case, the flow velocity, at different time steps from only the simulated distributions and the initial experimental values. In other words, given the distribution obtained using complex CFD that require elaborate numerical methods capable of solving the complicated differential equations given by the Navier-Stokes governing system, for a given time interval, it is possible to forecast the velocity distribution for future time steps, with estimated error given by the corresponding covariance matrix. In addition, the capability of predicting future values only with the simulated distribution and the fact that the associated error is computed as well, without even haven considered the experimental real data sets outlines the exceptional potential provided by the KF approach.

Once given the predicted distribution, the new observations are introduced in the algorithmic approach, improving the estimated values, in the filtering step.

Theorem B.3.7 (Kalman Filtering equations)**.** *The filtered estimates* $\boldsymbol{X}_{t|t} = P_t(\boldsymbol{X}_t)$ *and their covariance matrices* $\Sigma_{t|t} = E\left[(\boldsymbol{X}_t - \boldsymbol{X}_{t|t})(\boldsymbol{X}_t - \boldsymbol{X}_{t|t})^T\right]$ *are determined by the Kalman Filtering recursive equations:*

$$P_t(\boldsymbol{X}_t) = P_{t-1}(\boldsymbol{X}_t) + K(\boldsymbol{Y}_t - M_t\hat{\boldsymbol{X}}_t), \tag{B.20}$$

$$\Sigma_{t|t} = \Sigma_t - KM_t\Sigma_t^T, \tag{B.21}$$

$$K = \Sigma_t M_t^T (M_t\Sigma_t M_t^T + R_t)^{-1}. \tag{B.22}$$

Proof. From equation (B.17) it follows that

$$P_t\mathbf{X}_t = P_{t-1}\mathbf{X}_t + M\mathbf{I}_t,$$

where

$$\begin{aligned} M = E(\mathbf{X}_t\mathbf{I}_t^T) &= E\left[\mathbf{X}_t(G_t(\mathbf{X}_t - \hat{\mathbf{X}}_t) + \mathbf{W}_t)^T\right]\Delta_t^{-1} \\ &= E\left[\mathbf{X}_t(G_t(\mathbf{X}_t - \hat{\mathbf{X}}_t))^T\right]\Delta_t^{-1} = E\left[\mathbf{X}_t(\mathbf{X}_t - \hat{\mathbf{X}}_t)^T G_t^T\right]\Delta_t^{-1} \\ &= E\left[\mathbf{X}_t(\mathbf{X}_t - \hat{\mathbf{X}}_t)^T\right]G_t^T\Delta_t^{-1} = \Sigma_t G_t^T\Delta_t^{-1}. \end{aligned}$$

In order to establish the expression for the correlation matrices, we observe that

$$\mathbf{X}_t - P_{t-1}\mathbf{X}_t = \mathbf{X}_t - P_t\mathbf{X}_t + P_t\mathbf{X}_t - P_{t-1}\mathbf{X}_t = \mathbf{X}_t - P_t\mathbf{X}_t + M\mathbf{I}_t.$$

Using this expression, the equation given for the filtered estimates and the orthogonality of $\mathbf{X}_t - P_t\mathbf{X}_t$ and $M\mathbf{I}_t$, we get

$$\Sigma_t = \Sigma_{t|t} + \Sigma_t M_t^T\Delta_t^{-1}M_t\Sigma_t,$$

quod erat demonstrandum. □

Observe that the filtered mean given by the filtered recursive equation (B.20) could be expressed as a weighted average of the prior predicted estimate, $\hat{\mathbf{X}}_t$, and the observation vector, $\boldsymbol{Y}_t$. Similarly, the posterior precision given by (B.21) is the sum of the prior precision and the data precision (projected onto the state space).

Note that the aim of the Kalman filter is to find an improved analyzed estimate $\mathbf{X}_{t|t} = P_t\mathbf{X}_t$ of the true state $\mathbf{X}_t$. in order to find the distribution of the improved estimate observe that we assume that the errors are normally distributed, as specified in the definition of the state-space equations.

Theorem B.3.8. *When the errors are normally distributed, the expression of the improved analyzed estimate is given by the least squares estimate, which coincides with the maximum likelihood estimate, given by the expression* (B.20)*.*

Proof. We seek to find an improved linear estimator of the true state $\mathbf{X}_t$, given the observation vector $\mathbf{Y}_t$ and the linear predictor given by the Kalman prediction equations, that represents a linear predictor of the true state itself, and can be expressed as $\hat{\mathbf{X}}_t = \mathbf{X}_t + \varepsilon^p$, where we assume $\varepsilon^p \sim WN_N(0, \Sigma_p)$. By definition, note that $\Sigma_p = E\left[(\hat{\mathbf{X}}_t - \mathbf{X}_t)(\hat{\mathbf{X}}_t - \mathbf{X}_t)^T\right] = \Sigma_t$ represents the covariance matrix given by the Kalman prediction equations. Therefore, we may express the analyzed improved estimate as follows:

$$\mathbf{X}_{t|t} = \mathbf{X}_t + \varepsilon^a = \alpha\hat{\mathbf{X}}_t + \beta\mathbf{Y}_t. \tag{B.23}$$

Since we suppose gaussianity on the errors, we get $\varepsilon^a \sim WN_N(0, \Sigma_a)$. Observe that this represents the covariance matrix $\Sigma_a = \Sigma_{t|t}$ of the analyzed estimate. Therefore, $\mathbf{X}_{t|t}$ becomes an unbiased estimate of the true state, i.e. $\overline{\mathbf{X}_{t|t}} = \mathbf{X}_t$.
Using the expression of the estimate $\hat{\mathbf{X}}_t$ and the observation equation of the state-space representation in (B.23) we get:

$$\mathbf{X}_{t|t} = \mathbf{X}_t + \varepsilon^a = \alpha(\mathbf{X}_t + \varepsilon^p) + \beta(M\mathbf{X}_t + \mathbf{W}_t).$$

Taking expectations on both sides of the equation:

$$\overline{\mathbf{X}_{t|t}} = \mathbf{X}_t = (\alpha + \beta M)\mathbf{X}_t.$$

Observe that theoretically for each observation Y_t, we suppose we have the same linear combination of the components of the state vector $X_{j,t}$ for $j = 1, \ldots, k$. Therefore, conceptually, the observation equation is given in scalar dimensions for each t. Similarly, we can see that the observation matrix is interpreted as the vector $M^T \in \mathbb{R}^k$ that gives the linear combination of the state vector, which we assume that, theoretically, is the same for each t. As a result, we observe that we have $\alpha \in \mathbb{R}$, $\beta \in \mathbb{R}^k$.

Thus, we must have $\alpha I + \beta M = I$, where I denotes the identity matrix $k \times k$. Therefore, using $\alpha I = I - \beta M$ on the expression of the improved estimate we obtain:

$$\mathbf{X}_{t|t} = (I - \beta M)\hat{\mathbf{X}}_t + \beta Y_t = \hat{\mathbf{X}}_t + \beta(Y_t - M\hat{\mathbf{X}}_t). \tag{B.24}$$

Similarly, using the expression for the Kalman predictor and the observation equation on (B.23) we can find the expression for its error:

$$\begin{aligned}
&\mathbf{X}_{t|t} = \mathbf{X}_t + \varepsilon^a = \mathbf{X}_t + \varepsilon^p + \beta(M\mathbf{X}_t + W_t - M\mathbf{X}_t - M\varepsilon^p) \\
&\iff \varepsilon^a = \varepsilon^p + \beta(W_t - M\varepsilon^p).
\end{aligned}$$

Hence, we can compute the error covariance matrix, so we can minimize the error and obtain the least squares estimate of the true state that has minimum variance. By definition, the covariance matrix is given by:

$$\begin{aligned}
\Sigma_{t|t} &= \overline{(\mathbf{X}_t - \mathbf{X}_{t|t})(\mathbf{X}_t - \mathbf{X}_{t|t})^T} = \overline{\varepsilon^a(\varepsilon^a)^T} \\
&= \overline{(\varepsilon^p + \beta(W_t - M\varepsilon^p))(\varepsilon^p + \beta(W_t - M\varepsilon^p))^T} \\
&= \overline{\varepsilon^p(\varepsilon^p)^T} + \beta\overline{\varepsilon^p(W_t - M\varepsilon^p)^T} + \beta\overline{(W_t - M\varepsilon^p)(\varepsilon^p)^T} + \beta^2\overline{(W_t - M\varepsilon^p)(W_t - M\varepsilon^p)^T} \\
&= \Sigma_p - \beta\overline{\varepsilon^p(\varepsilon^p)^T}M^T - \beta M\overline{\varepsilon^p(\varepsilon^p)^T} + \beta^2(\overline{W_tW_t^T} + M\overline{\varepsilon^p(\varepsilon^p)^T}M^T),
\end{aligned}$$

where we used the uncorrelation between the errors, i.e. $\overline{W_t(\varepsilon^p)^T} = 0$, and the fact that M is the observation matrix and therefore it is a deterministic matrix so the expectation operator does not change its value. Taking the expectations that are left, according to the definition of the corresponding errors we obtain:

$$\Sigma_{t|t} = \Sigma_p - \beta\Sigma_pM^T - \beta M\Sigma_p + \beta^2(R + M\Sigma_pM^T).$$

Note that if we denote $r = M\Sigma_p$, it is direct to see that $r^T = \Sigma_p^T M^T = \Sigma_p M^T$, since $M^T \in \mathbb{R}^k$. Thus, we get:

$$\Sigma_{t|t} = \Sigma_p - 2\beta r^T + \beta^2(R + M\Sigma_pM^T).$$

If we derive to optimize:

$$\frac{\partial \Sigma_{t|t}}{\partial \beta} = -2r^T + 2\beta(R + M\Sigma_p M^T) = 0$$
$$\iff \beta = r^T(\Sigma_{t|t} + M\Sigma_p M^T)^{-1}.$$

Then, the covariance matrix that gives the minimum[3] variance is given by:

$$\begin{aligned}\Sigma_{t|t} &= \Sigma_p - 2r^T(R + M\Sigma_p M^T)^{-1}r + r^T(R + M\Sigma_p M^T)^{-1}r \\ &= \Sigma_p - r^T(R + M\Sigma_p M^T)^{-1}r. \end{aligned} \tag{B.25}$$

Using expression (B.24) we get the least squares solution $\mathbf{X}_{t|t}$ that gives the minimum variance estimate:

$$\mathbf{X}_{t|t} = \hat{\mathbf{X}}_t + r^T(R + M\Sigma_p M^T)^{-1}(Y_t - M\hat{\mathbf{X}}_t). \tag{B.26}$$

Consequently, expressions (B.25) and (B.26) give us the recursive equations that represent the Kalman filter equations, proving that the improved analyzed estimate $\mathbf{X}_{t|t}$ corresponds to the least squares estimate, as we wanted to prove.

Using now the Bayesian formulation and the assumptions that the errors are normally distributed we may prove that the maximum likelihood estimate also gives the analyzed estimate given by the Kalman filtering equations.
Given a density function $f(\mathbf{X})$ for the first-guess estimate $\hat{\mathbf{X}}_t$, and a likelihood function $f(Y|\mathbf{X})$ for the measurement Y_t, then from the Bayes' theorem we have:

$$f(\mathbf{X}|Y) \propto f(\mathbf{X})f(Y|\mathbf{X}).$$

Given the state-space equations and the representation for the first-guess estimate in terms of the true state $\mathbf{X}_t$ which is deterministic, it is direct to see that if $\varepsilon^p \sim N_k(0, \Sigma_t)$ and $W \sim N_k(0, R)$ then $\hat{\mathbf{X}}_t$ and Y_t are normally distributed, $\hat{\mathbf{X}}_t \sim N_k(\mathbf{X}, \Sigma_t)$ and $Y_t \sim N(M\mathbf{X}, R)$. Consequently, their density functions are given by the multivariant Gaussian density function:

$$f(\mathbf{X}) = \frac{1}{|\Sigma_t|^{1/2}(2\pi)^{k/2}} e^{-\frac{1}{2}(\hat{\mathbf{X}}_t - \mathbf{X})^T(\Sigma_t)^{-1}(\hat{\mathbf{X}}_t - \mathbf{X})},$$
$$f(Y_t|\mathbf{X}) = \frac{1}{|R|^{1/2}(2\pi)^{m/2}} e^{-\frac{1}{2}(Y_t - M\mathbf{X})^T(R)^{-1}(Y_t - M\mathbf{X})}.$$

Note that, given the assumptions over the scalar observation equation, here $m = 1$ simplifying the following calculations. Therefore, the likelihood function for the firts-guess estimate can be expressed in terms of the Mahalanobis distances described above, $f(X|Y) \propto e^{-\frac{1}{2}\mathcal{J}[X]}$ where we define the variational function as:

$$\mathcal{J}[\mathbf{X}] = (\hat{\mathbf{X}}_t - \mathbf{X})^T(\Sigma_t)^{-1}(\hat{\mathbf{X}}_t - \mathbf{X}) + (Y_t - M\mathbf{X})^T(R)^{-1}(Y_t - M\mathbf{X}).$$

Observe that the minimum of $\mathcal{J}[\mathbf{X}]$ gives the maximum of $f(\mathbf{X}|Y_t)$. The minimum of the variational function is found from:

$$\frac{\partial \mathcal{J}}{\partial \mathbf{X}}[\mathbf{X}] = -2(\Sigma_t)^{-1}(\hat{\mathbf{X}}_t - \mathbf{X}) - 2(R)^{-1}M^T(Y_t - M\mathbf{X}) = 0,$$
$$\iff (\hat{\mathbf{X}}_t - \mathbf{X}) = \Sigma_t(R)^{-1}M^T(Y_t - M\mathbf{X}).$$

[3] It is direct to see that it is a minimum, by computing the second derivative and from the fact that covariance matrices are definite positive.

The corresponding calculus derives from the algebraic derivation of the vectorial expressions:

$$g(a) = u^T a, \quad \frac{\partial g}{\partial a}(a) = u, \text{ where } a, u \in \mathbb{R}^k,$$
$$g(a) = a^T M a, \quad \frac{\partial g}{\partial a}(a) = 2Ma, \text{ where } M \in \mathcal{M}_k(\mathbb{R}).$$

Keeping in mind that $R \in \mathbb{R}$, and $(Y_t - M\mathbf{X}) \in \mathbb{R}$, we can solve this equation in order to find $\mathbf{X}$ that maximizes the likelihood as wanted:

$$\begin{aligned}
&(R+M\Sigma_t M^T)\mathbf{X} = R\hat{\mathbf{X}}_t + \Sigma_t M^T Y \iff \mathbf{X} = (R + M\Sigma_t M^T)^{-1}(R\hat{\mathbf{X}}_t + \Sigma_t M^T Y),\\
\iff\ & \mathbf{X} = (R + M\Sigma_t M^T)^{-1}\left(R\hat{\mathbf{X}}_t + (R + M\Sigma_t M^T)\hat{\mathbf{X}}_t - (R + M\Sigma_t M^T)\hat{\mathbf{X}}_t + \Sigma_t M^T Y\right),\\
\iff\ & \mathbf{X} = \hat{\mathbf{X}}_t + (R + M\Sigma_t M^T)^{-1}\left(\Sigma_t M^T Y - M\Sigma_t M^T \hat{\mathbf{X}}_t\right),\\
\iff\ & \mathbf{X} = \hat{\mathbf{X}}_t + (R + M\Sigma_t M^T)^{-1}\Sigma_t M^T\left(Y - M\hat{\mathbf{X}}_t\right).
\end{aligned}$$

For the last equality, it is used that $M\Sigma_t M^T \in \mathbb{R}$ and therefore commutes.[4]

□

B.4 Ensemble Kalman filter

If a model forecast $\hat{\mathbf{X}}_t$ and the forecast error covariance Σ_t are known at a time t, where we have available observations $\{\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_t\}$, with a measurement error covariance matrix $\mathbf{W}_t$, it is possible to calculate an improved analysis $\mathbf{X}_{t|t}$, with its covariance matrix $\Sigma_{t|t}$. The problem gets more complex when we do not have the true state, for instance, when studying not analytical solutions of the Navier-Stokes equations, and therefore we have to use the best estimate of $\hat{\mathbf{X}}_t$, which is given by the ensemble mean $\overline{\mathbf{X}}$. In practice, the EnKF is also considered with high dimensional problems, when the state dimension considerably difficults the method implementation. In the Kalman filter, the error covariance matrices for the predicted and the analyzed estimate are given by:

$$\Sigma_t = \overline{(\hat{\mathbf{X}}_t - \mathbf{X}_t)(\hat{\mathbf{X}}_t - \mathbf{X}_t)^T},$$
$$\Sigma_{t|t} = \overline{(\mathbf{X}_{t|t} - \mathbf{X})(\mathbf{X}_{t|t} - \mathbf{X}_t)^T}.$$

Therefore, in the Ensemble Kalman filter approach we define the ensemble covariance matrices around the ensemble mean:

$$\Sigma_t^e = \overline{(\hat{\mathbf{X}}_t - \overline{\hat{\mathbf{X}}_t})(\hat{\mathbf{X}}_t - \overline{\hat{\mathbf{X}}_t})^T},$$
$$\Sigma_{t|t}^e = \overline{(\mathbf{X}_{t|t} - \overline{\mathbf{X}_{t|t}})(\mathbf{X}_{t|t} - \overline{\mathbf{X}_{t|t}})^T}.$$

Consequently, the analysis step in the Ensemble Kalman filter (EnKF) consists on adapting these definitions to the sequential equations described for the KF as shown, [5]:

$$\overline{\mathbf{X}_{t|t}} = \overline{\hat{\mathbf{X}}_t} + K^e(\overline{\mathbf{Y}_t} - M_t\overline{\hat{\mathbf{X}}_t}), \tag{B.27}$$
$$\Sigma_{t|t}^e = \Sigma_t^e - K^e M_t (\Sigma_t^e)^T, \tag{B.28}$$
$$K^e = \Sigma_t^e M_t^T (M_t \Sigma_t^e M_t^T + R_t)^{-1}. \tag{B.29}$$

[4] This proof is a generalization from the uni-dimensional proof from [5].

In contrast to the standard KF which works with the entire distribution of the state explicitly, the EnKF stores, propagates and updates an ensemble of the vectors that approximate the state distribution. Computationally, the EnKF provides a more efficient recursive method to inference high-dimensional nonlinear and non-Gaussian physical systems. The key difference between the EnKF and other sequential algorithms is the use of a linear updating rule that converts the prior ensemble to a posterior ensemble after each observation.

The state distribution is represented by a sample or ensemble from the distribution, resulting in a reduction of the initial dimension of the data sets. Specifically, assume that the ensemble $\{\mathbf{X}^{(1)}_{t-1|t-1}, \ldots, \mathbf{X}^{(N)}_{t-1|t-1}\}$ is a sample from the filtering distribution at time $t-1$. This ensamble is propagated forward through time and updated when new data become available through the forecast and the update steps.

B.5 Computational implementation of the DA methods

We seek to find the joint probability density functions (*pdf*) of the parameters and model state, given a set of measurements and a dynamical model with known uncertainties. We will define a physical model based on a wind flow in a cylindrical pipe, with associated initial and boundary conditions on the spatial domain.
In this study, the observation vector, $\boldsymbol{Y}_t$, is given by the velocity measured experimentally at a set of p points of the wind flow domain. Meanwhile, the state vector, $\boldsymbol{X}_t$, represents the velocity flow obtained through simulations or numerical predictions at a set of k points of the domain. The prediction matrices described on the equations (B.2), F_t, are fixed over time, since they represent the state variable time evolution, determined through numerical resolution of the RANS equations, in our case. Thus, for each time-step the physical model describing the system is described by the same governing equations. Similarly, the observation matrices (B.1), determine an equivalency between the measured values and the simulated distribution.
On the other hand, the covariance matrices of the measured and simulated errors ($\mathbf{W}_t, \mathbf{V}_t$, respectively) are considered to be constant over time, and diagonal, assuming uncorrelation. The exact error magnitude is given by the experimental and simulated data set standard deviation.

The boundary conditions are given by zero velocity at the upper and lower pipe walls, as expected from a parallel steady flow, atmospheric pressure at the outlet and a white noise distributed incoming velocity through the inlet (entrance region at the pipe's edge).

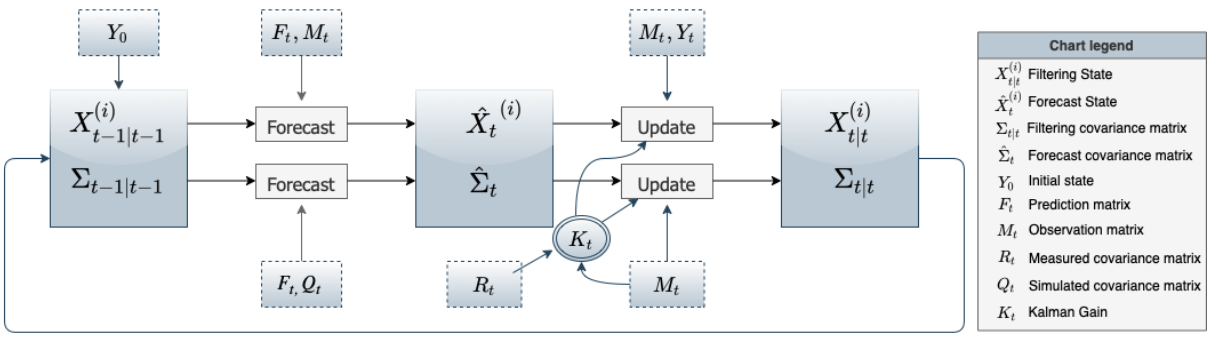

Figure B.1: Ensemble Kalman Filter information flow chart diagram. Here $i \in \{1, \ldots, N\}$, where N either corresponds to the sample dimension in the KF algorithm, or to the ensemble size, in the EnKF.

Computationally, the KF forecasting step is implemented over the simulated data (true state, $\boldsymbol{X}_t$), and the update step includes the experimental data set (observation vector, $\boldsymbol{Y}_t$). The KF algorithm idea is represented on Figure B.1, where the predict and update steps are implemented over the total true state. Note that the forecast and update state is implemented over the state variable itself and over the covariance matrix equivalently on each time step.

Similarly, the EnKF is implemented in order to contrast results, as described on [11]. The EnKF only requires storing and operating on N vectors (ensemble members) of length k and the calculation of the Kalman gain as described by the equation (B.29) using the sample covariance matrix.
As aforementioned, the EnKF algorithm starts with an initial ensemble $\{\mathbf{X}_0^{(1)}, \ldots, \mathbf{X}_0^{(N)}\}$, given by the initial state estimates. Then, at each time step, given the ensemble $\{\mathbf{X}_{t-1|t-1}^{(1)}, \ldots, \mathbf{X}_{t-1|t-1}^{(N)}\}$ of draws from the filtering distribution at time $t-1$, the stochastic EnKF[5] consists of the two steps implemented over each ensemble's member for $i = 1, \ldots, N$. Using the corresponding state-space representation of the simulated data, the forecast step involves drawing $\boldsymbol{w}_t^{(i)} \sim WN(0, Q_t)$ and calculate $\hat{\mathbf{X}}_t^{(i)} = F_t \mathbf{X}_{t-1|t-1}^{(i)} + \boldsymbol{w}_t^{(i)}$, where the transition matrix has been computed using the time series model that approximates the simulated data, representing the true state distribution. Following up, the update step consists of drawing $\boldsymbol{v}_t^{(i)} \sim WN(0, R_t)$ and calculating the filtered state $\mathbf{X}_{t|t}^{(i)} = \hat{\mathbf{X}}_t^{(i)} + K_t^e(\boldsymbol{y}_t + \boldsymbol{v}_t^{(i)} - M_t \hat{\mathbf{X}}_t^{(i)})$. An illustrative EnKF algorithm is represented on Figure B.1.

[5]The EnKF update step could be carried out using deterministic updates, in which the ensemble is approximated by deterministically shifting the prior ensemble, without relying on simulated or perturbed observations.

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